Explainable AI (XAI)

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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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Data-driven Intelligence & Learning Lab

AIML is everywhere

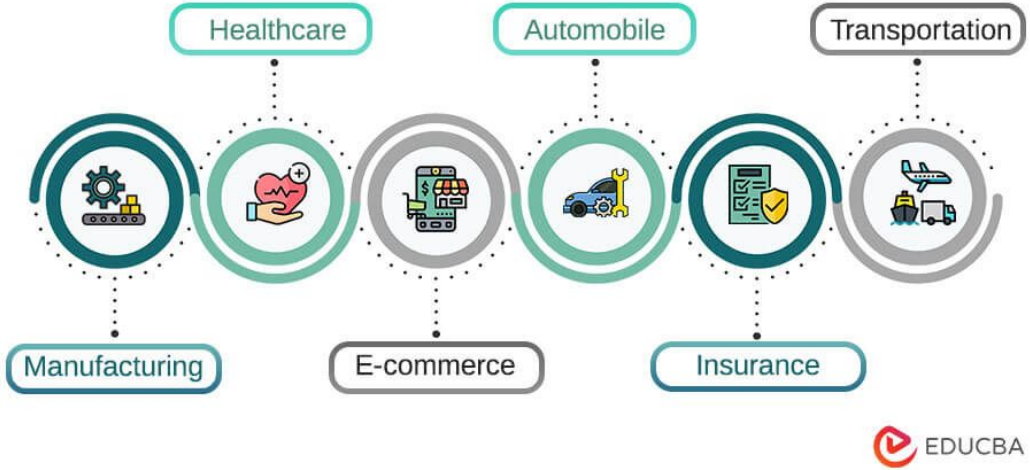


Figure credits: EDUCBA

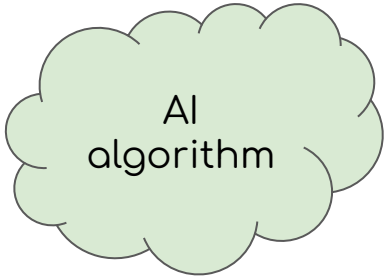
Let's consider an example

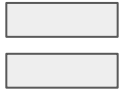
- You built an AI system to detect fraudulent transactions

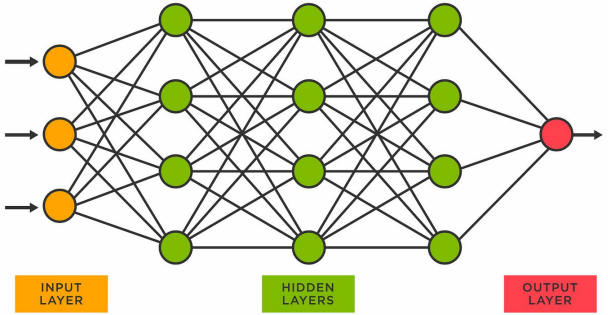
AI algorithm

Let's consider an example

- You built an AI system to detect fraudulent transactions







Let's consider an example

- Somebody paid 10000 INR in a coffee shop
- 'Fraudulent' transaction?
  - Why?

Can the model help us understand Why?

AI models are sophisticated

- Sometimes, the designers themselves can't fully understand these sophisticated models

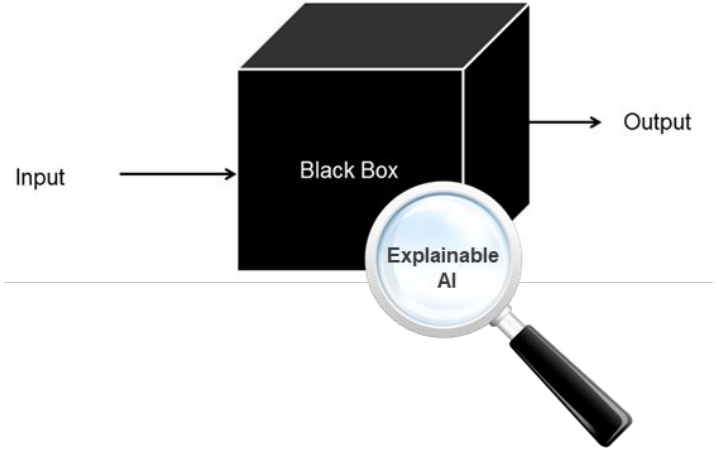


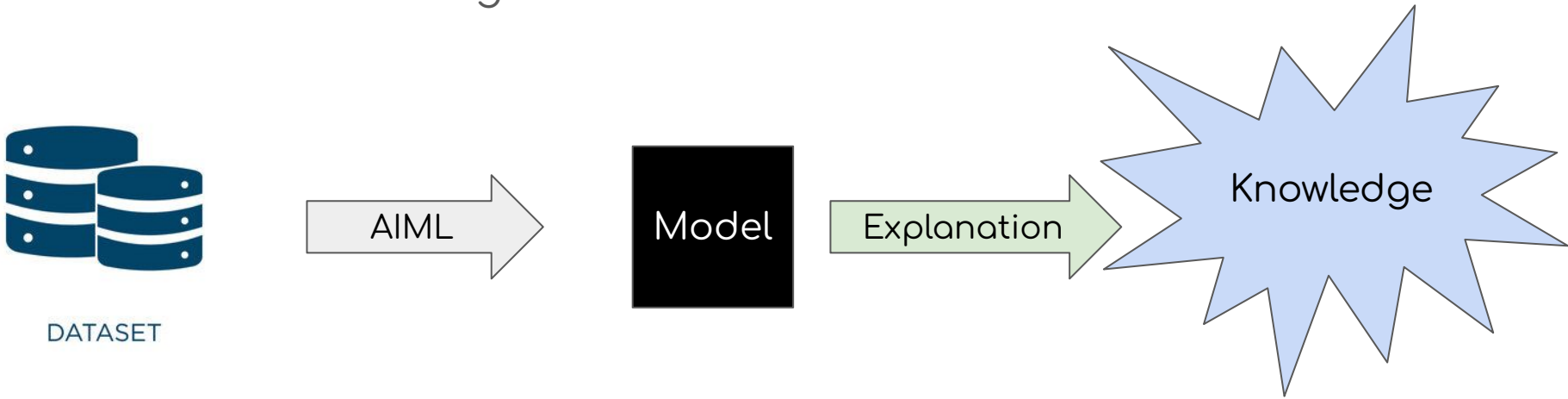
Figure credits: Worldline

Explainable AI

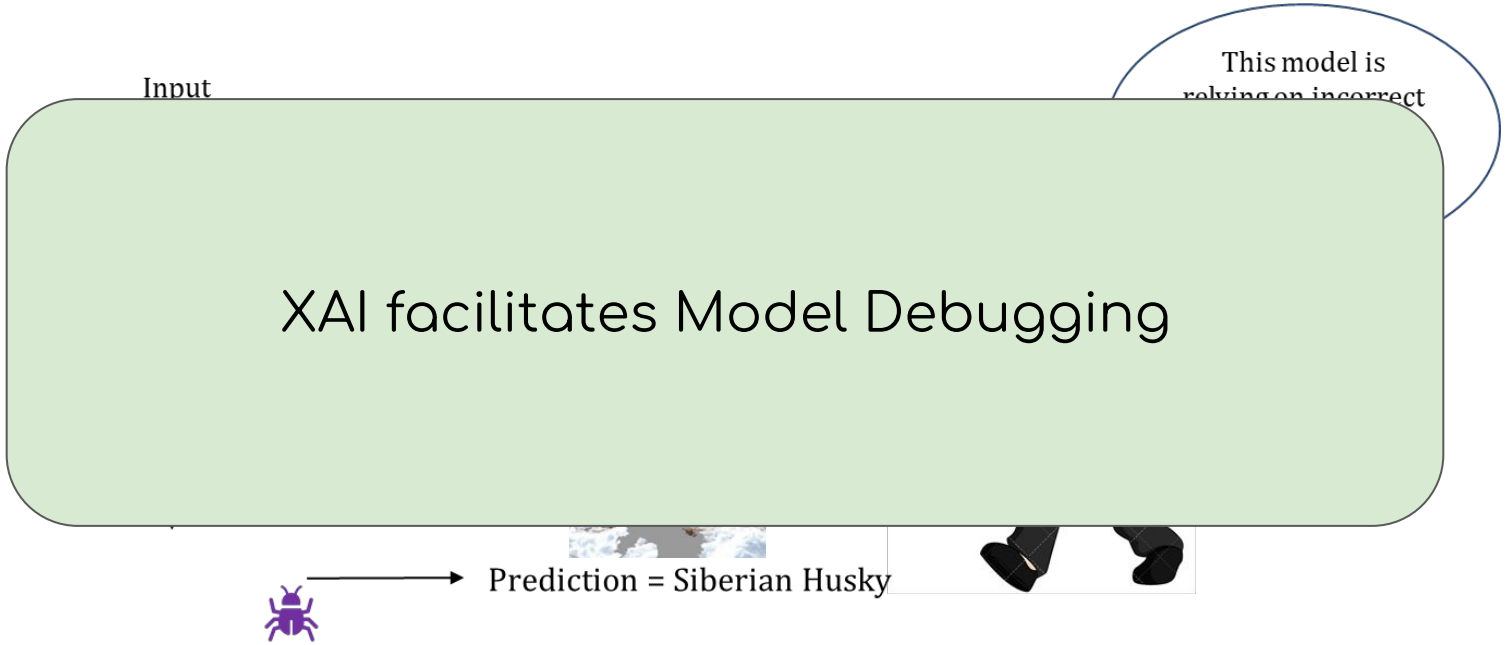
Collection of techniques to help us understand how the ML models produce inference

Explainable AI

- Science → Knowledge

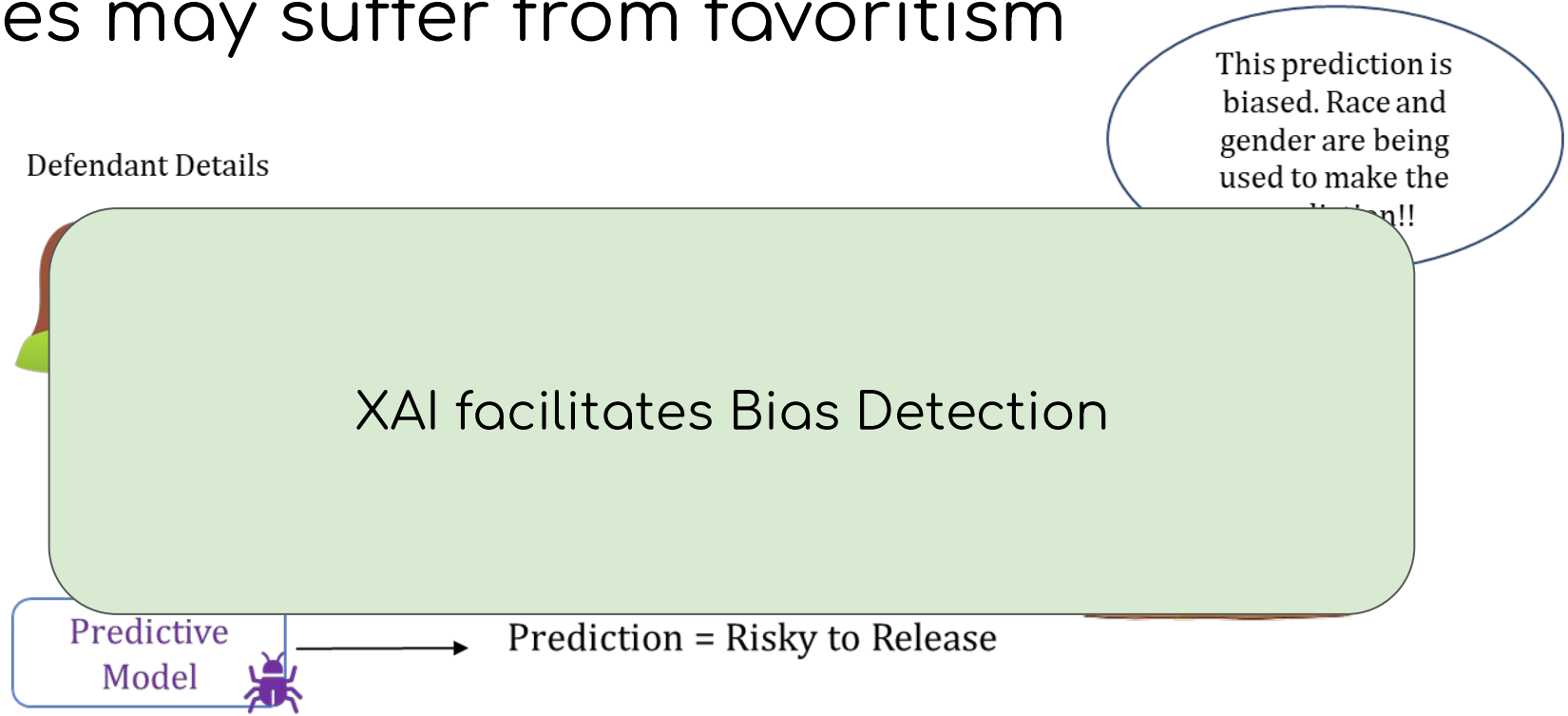


Not all stories are exciting



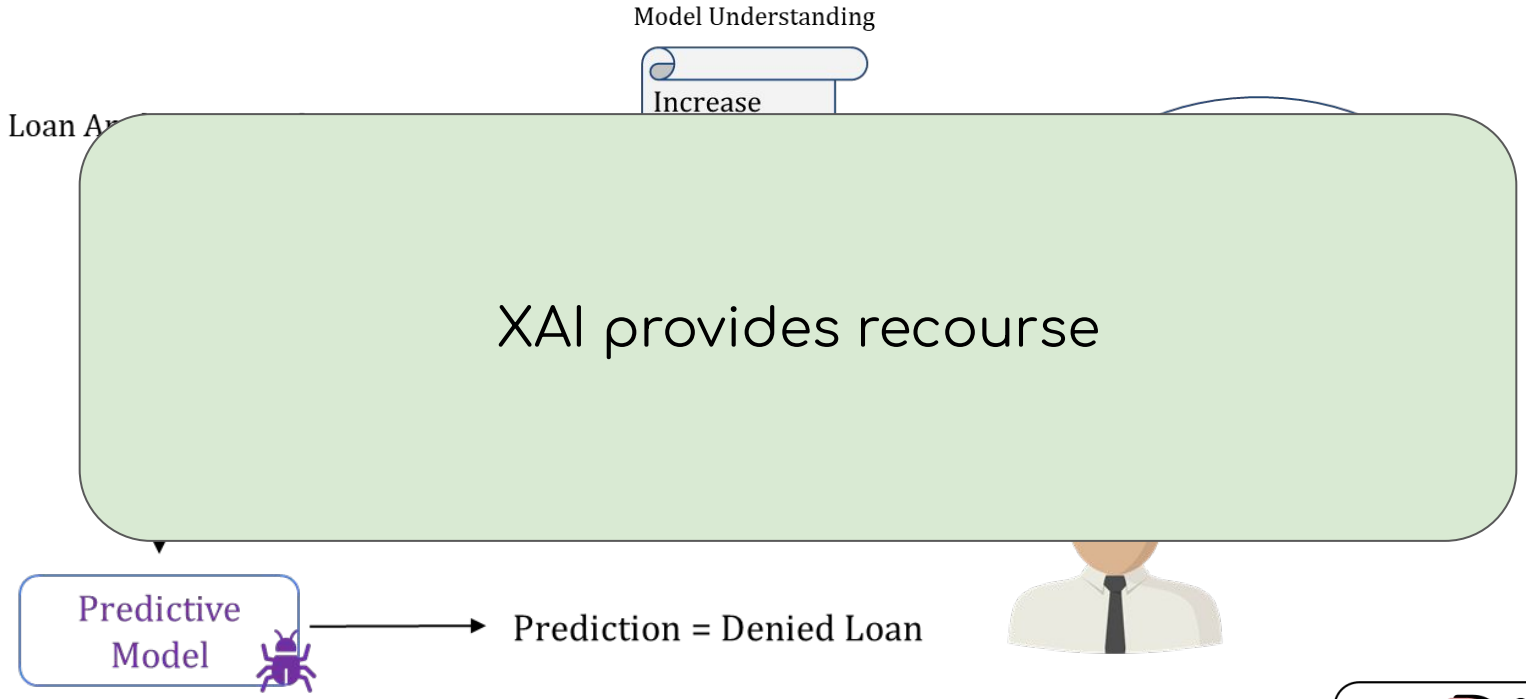
XAI facilitates Model Debugging

Stories may suffer from favoritism



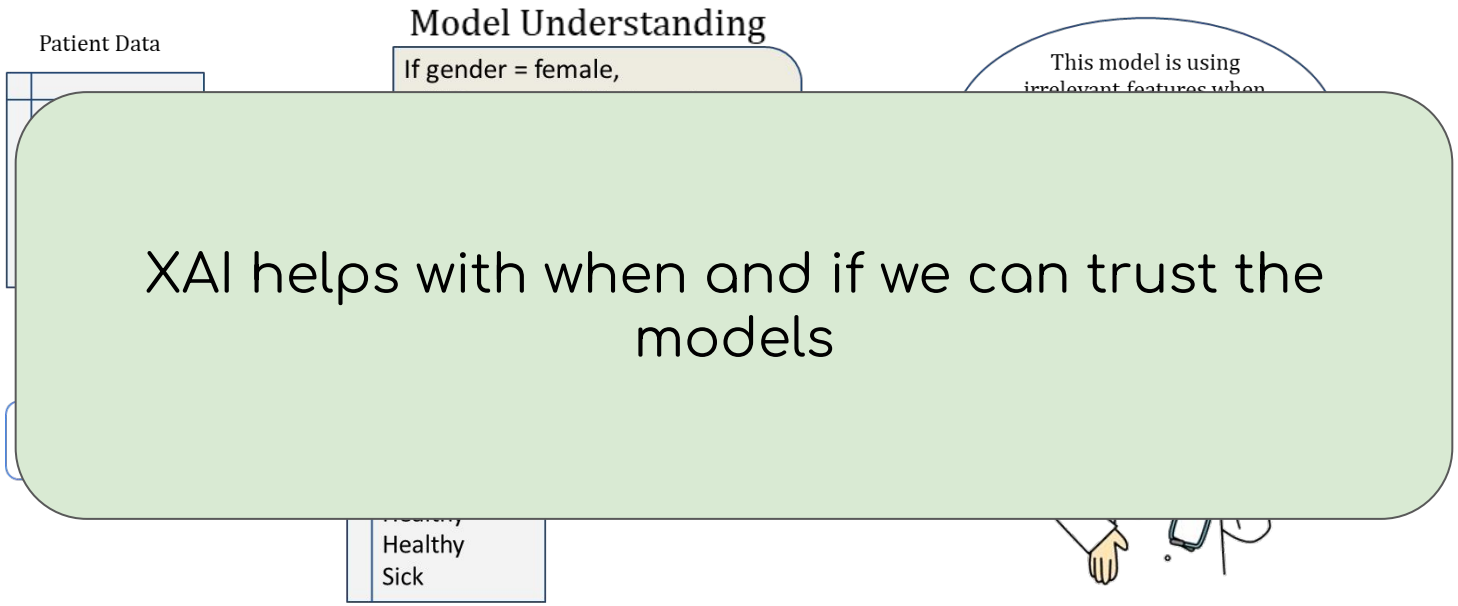
Slide credit Prof. Hima Lakkaraju

Stories may guide the users



Slide credit Prof. Hima Lakkaraju

Can pick that ‘good’ story



XAI helps with when and if we can trust the models

Slide credit Prof. Hima Lakkaraju

Why Explainability?

- Developers
  - Debug and Improve the models → Leads to new insights

Why Explainability?

- Regulators
  - Bias detection → Ensure the fairness
  - If-and-when to trust the models

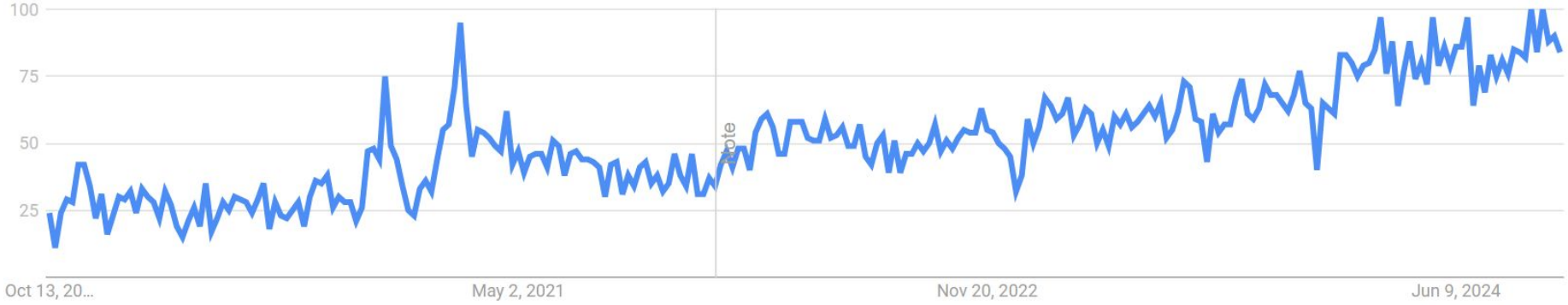
Why Explainability?

- End-users
  - Justify the inference (respond to the queries & build trust)
  - Provide recourse

Why Explainability?

More ML dependence → more need for explanations

Growing Emphasis on Explainability



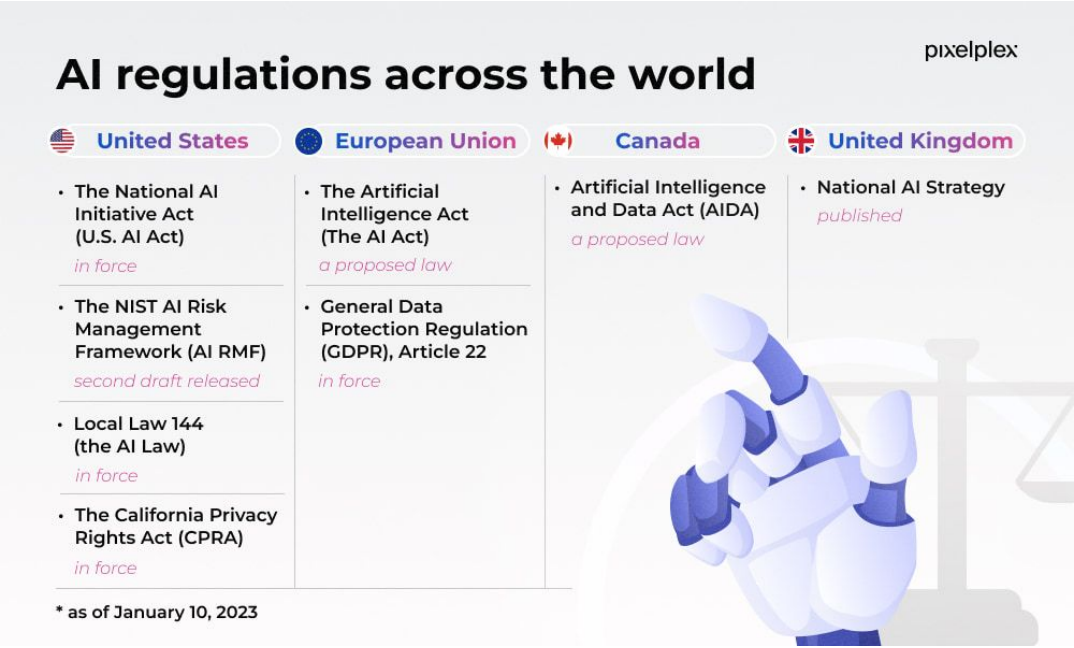
Google Trends

Accuracy alone may be insufficient

- Test sets may not be complete
- Auxiliary criteria for model evaluation
  - Safety
  - Fairness
  - Right to explanations
  - ....

Doshi-Velez and Kim 2017

Why Explainability?



What is Explainability/Interpretability?

Explainability/Interpretability

- Not simple to define mathematically

Interpretability

- Degree to which a human can understand the cause of a decision (Miller 2017)
- Degree to which a human can consistently predict the model's results (Kim et al. 2016)

Interpretability

- However
  - What constitutes an explanation?
  - What makes one better than another?

Interpretability

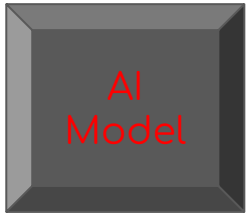
- Extraction of relevant knowledge from a model concerning relationships contained in the data or learned by the model (Murdoch et al. 2019)

Explanation?

- (Human) Interpretable description of the model behavior

Explanation

Understandable



Explanation

Faithful

User

Explanation

Explanation

AI Model

Model parameters?

Example predictions?

Summarize with a program/rule/tree?

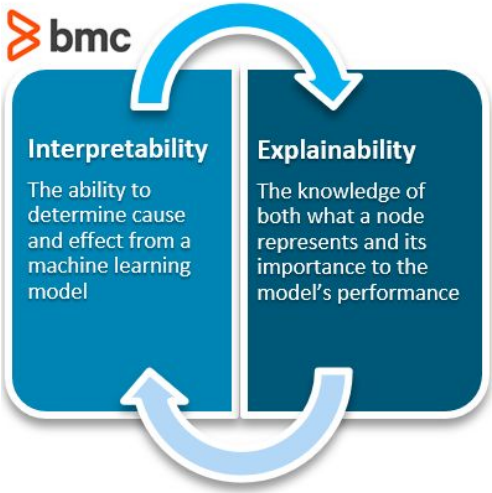
Most important features?

Describe how to flip the prediction?

User

Interpretability vs. Explainability

What's the hypothesis of the ML model?

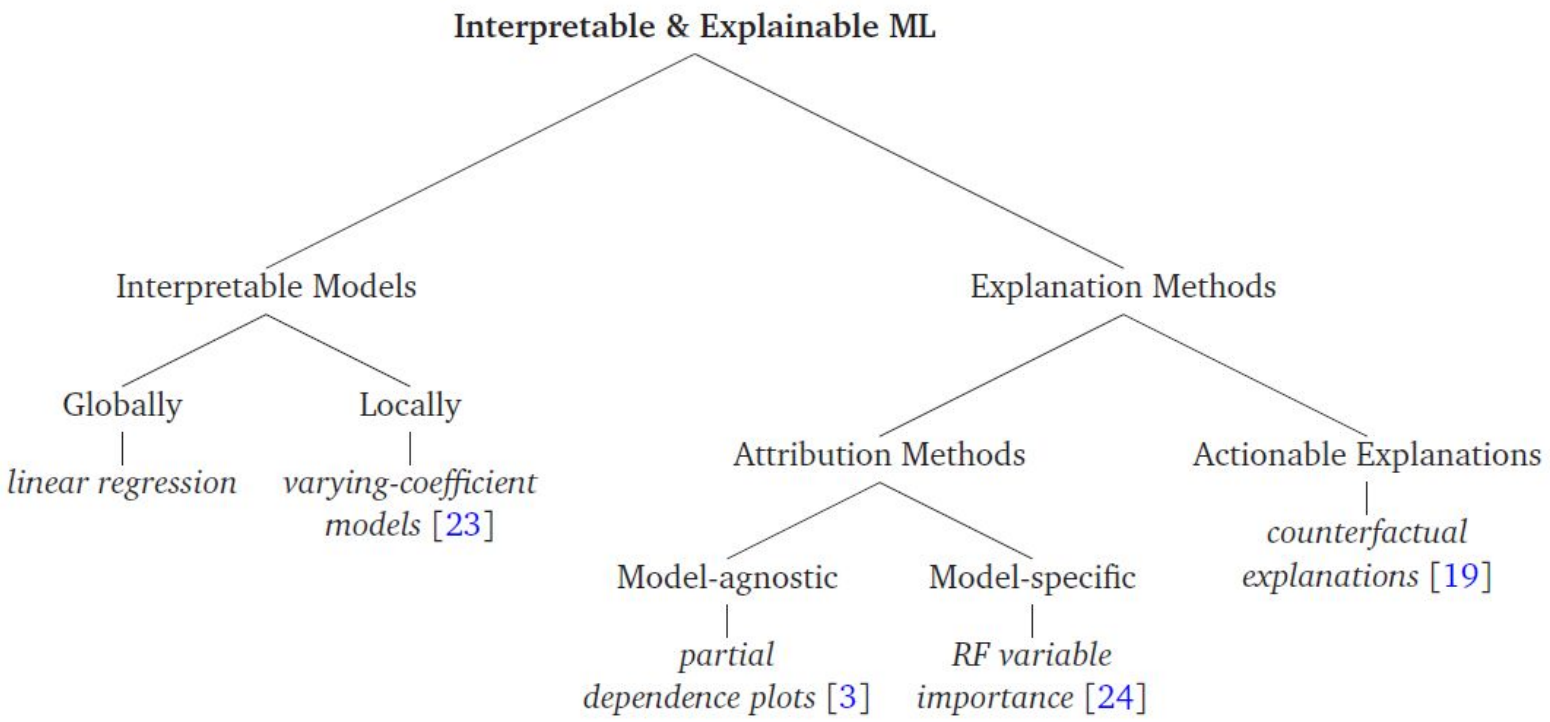


How does the components of the ML model support/explain it?

Figure by Jonathan Johnson

Interpretability vs. Explainability

- Interpretable ML focuses on designing models that are inherently interpretable
- Explainable ML tries to provide post hoc explanations for existing black box models

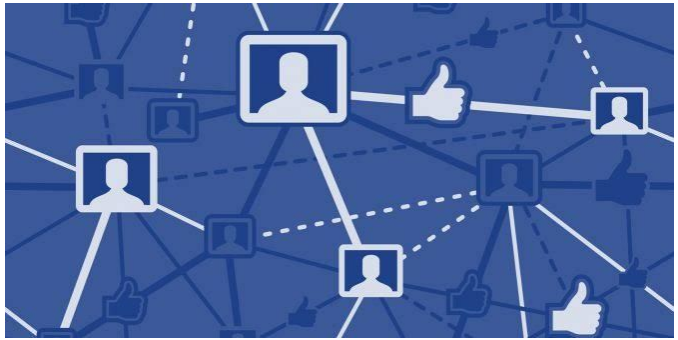


From Marcinkevi˘cs[2], based on the survey by Carvalho et al. [3]

When Explainability?

Need emphasis always?

- Model has no significant impact (non-critical applications)



Need emphasis always?

- Problem is well studied

Need emphasis always?

- High-level predictions
  - Marketing strategies

Is all well?

- Explainability may lead to ‘model manipulation’

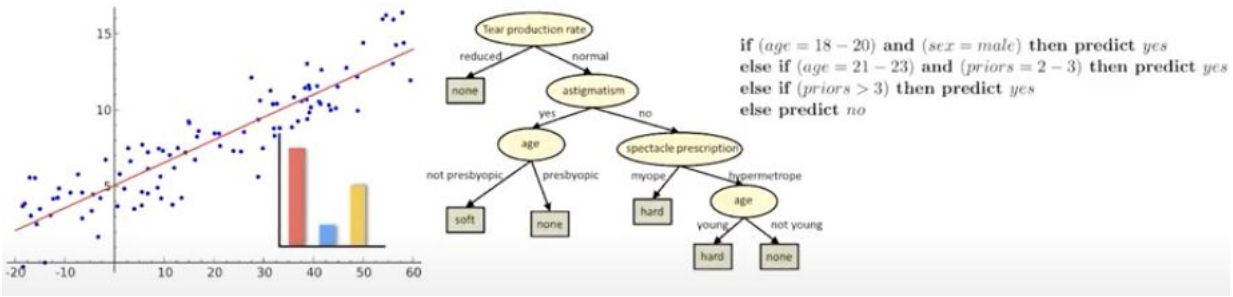
Is all well?

- False sense of trust
  - Accept erroneous predictions by an interpretable model over predictions of a black-box

How to achieve model understanding?

Achieving model understanding

1. Build inherently interpretable models (Model-based)
   a. Linear models
   b. Shallow decision trees, etc



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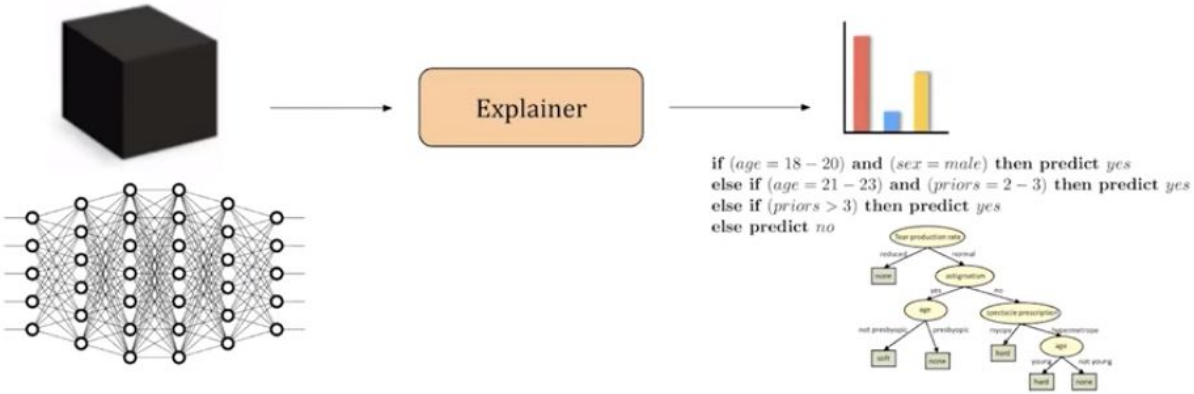
Achieving model understanding

2. Explain pre-built models in post-hoc manner



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When to go for what?

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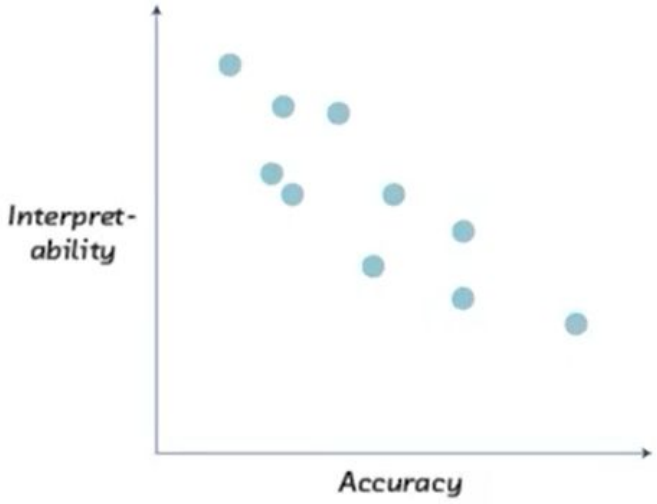
Accuracy vs interpretability



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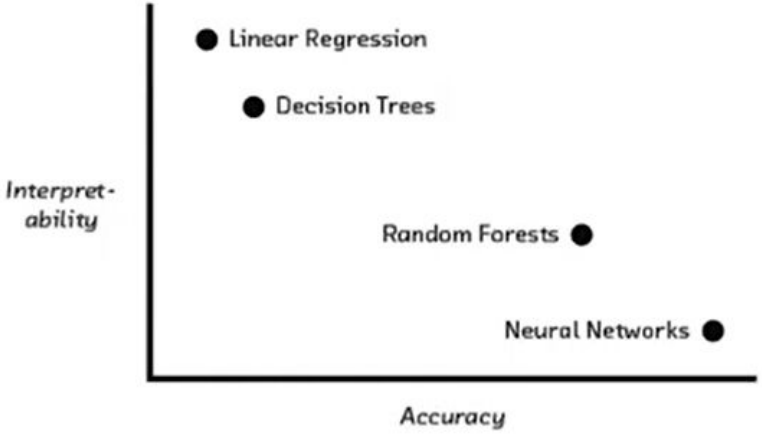
Accuracy vs interpretability



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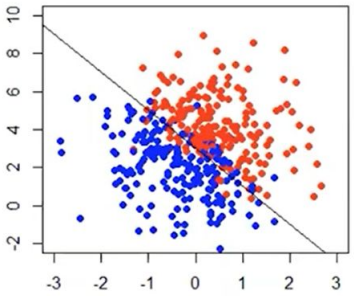
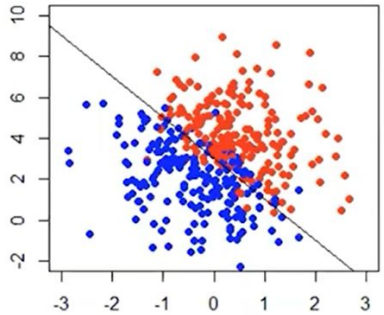
Interpretable models vs post-hoc explanations



can build interpretable + accurate models

Interpretable models vs post-hoc explanations



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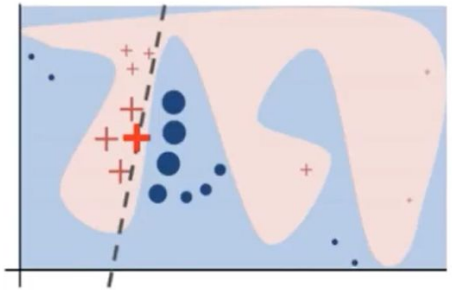

complex models might achieve higher accuracy

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Interpretable models vs post-hoc explanations

- Sometimes, we only have the trained model (a black box)
- You don't have the training data to build the model from scratch
- → post-hoc explanations

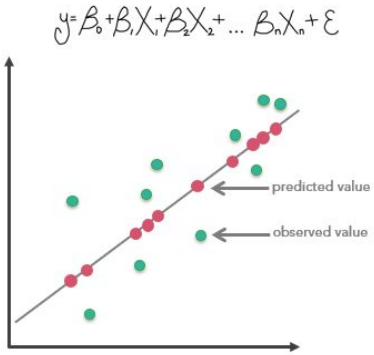
In other words

- If we can build an interpretable model with adequate accuracy → go for it
- Otherwise resort to post-hoc explanations for the complex models

Inherently Interpretable models

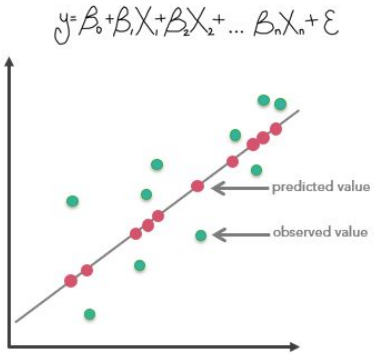
Linear Regression

- Fits a linear model to the data
- $Y = \square_0 + \square_1 x_1 + \ldots\ldots + \square_n x_n$
- Parameters are learned using a loss function and a training data



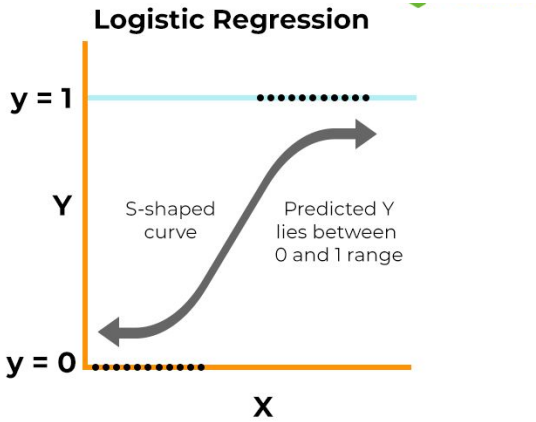
Linear Regression: Interpretability

- $Y = \square_0 + \square_1 x_1 + \ldots\ldots + \square_n x_n$
- Impact of an i/p variable x_i can be interpreted as $\square_i$
- **Changing it by 1 → $\square_i$ change in Y** (attention to the sign!)



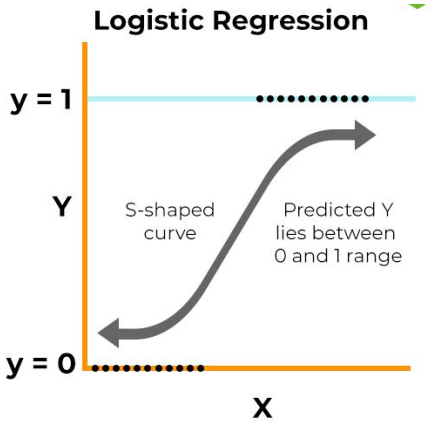
Logistic Regression

- Extension of linear regression to classification



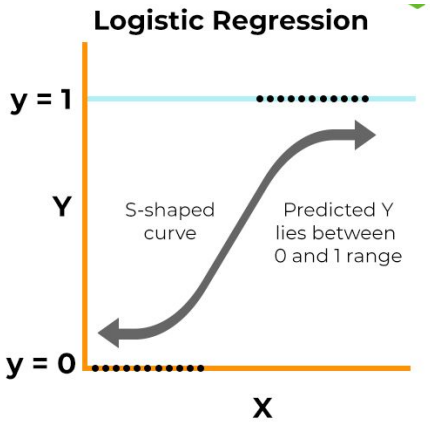
Logistic Regression

- $Y = \sigma(\square_0 + \square_1 x_1 + \ldots\ldots + \square_n x_n)$
- Parameters are learned during the optimization over a training dataset



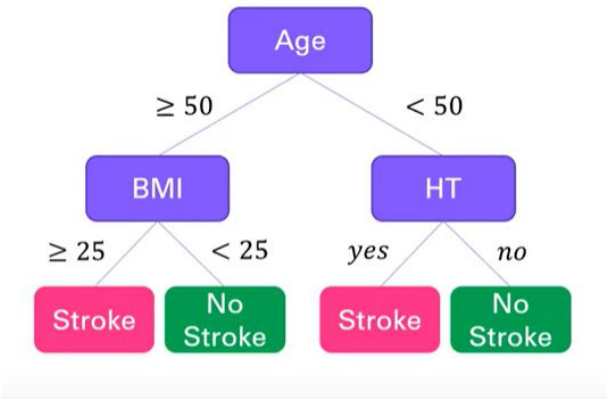
Logistic Regression: Interpretability

- $Y = \sigma(\square_0 + \square_1 x_1 + \ldots\ldots + \square_n x_n)$
- Impact of an i/p variable x_i can be interpreted as $e^{\square i}$



Decision Trees

- For the ‘stroke’ problem, one of the hypotheses can be



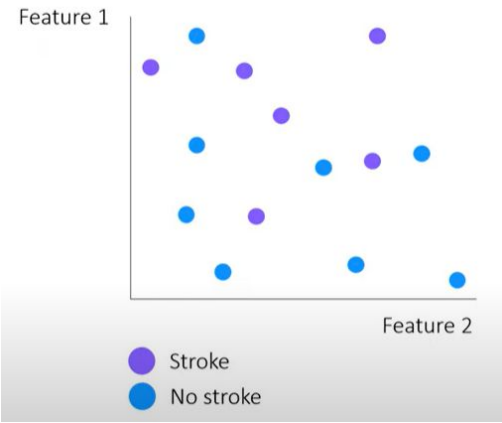
Post-hoc explanations

LIME

- Local Interpretable Model-agnostic Explanations

LIME

- Let's consider a binary classification



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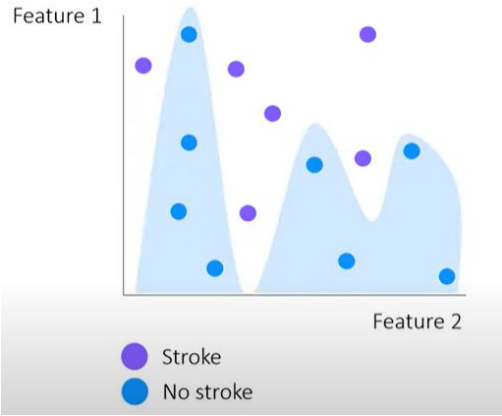

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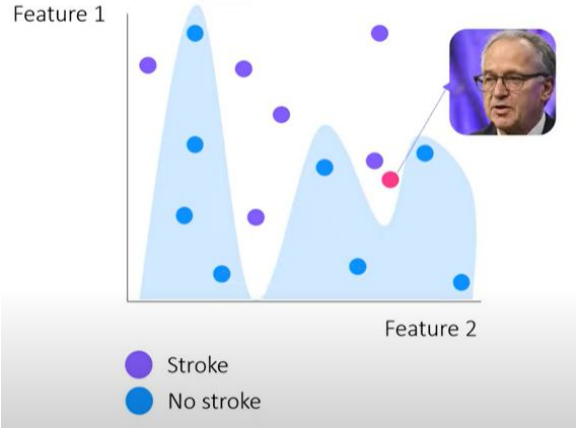

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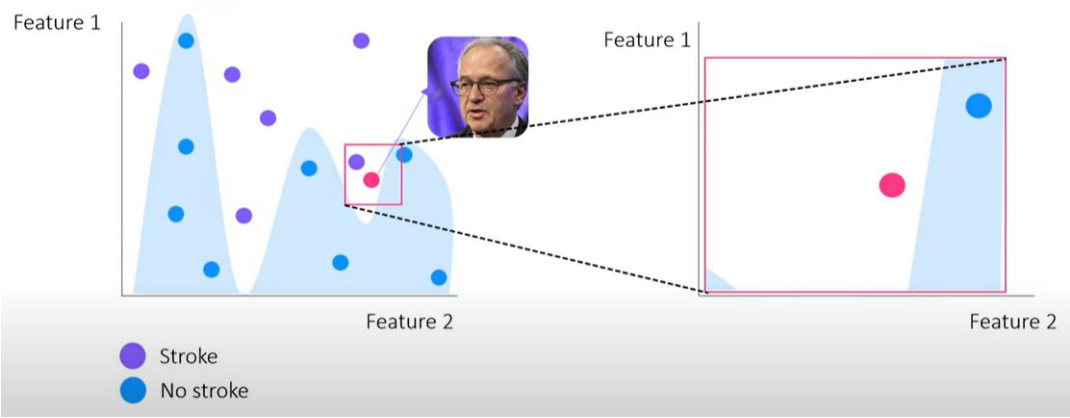

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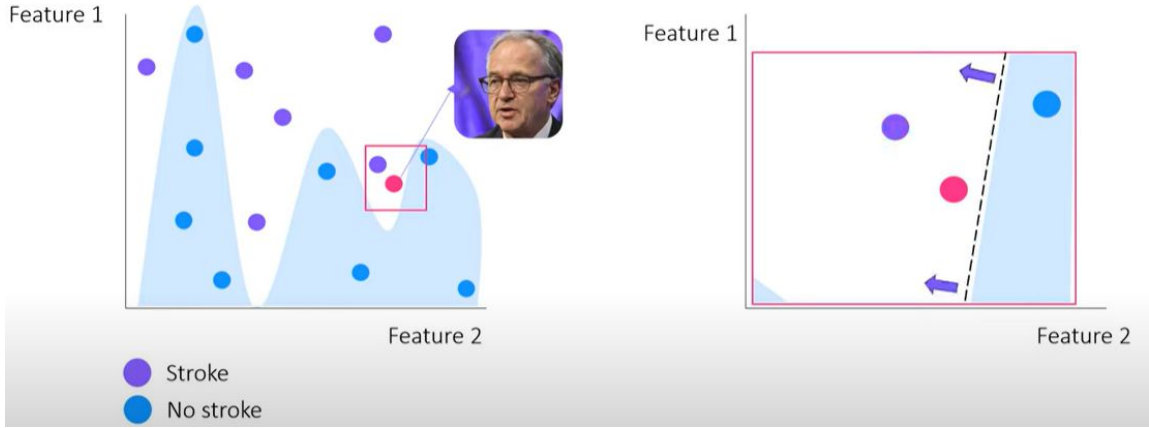
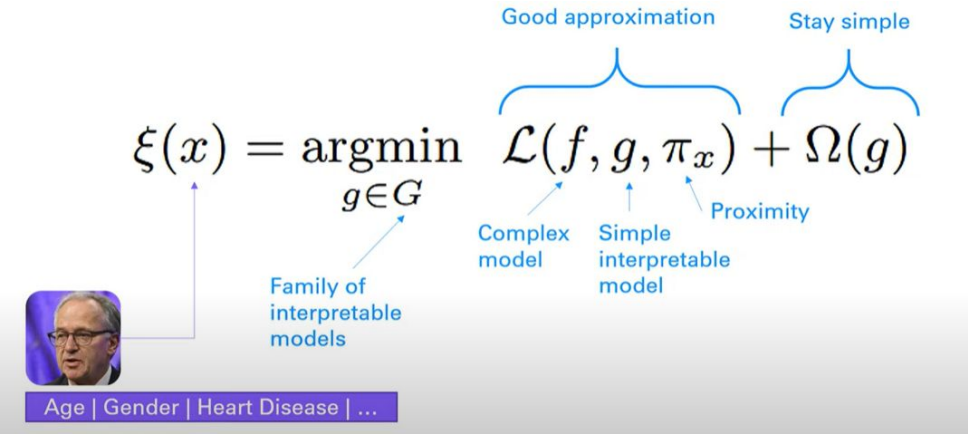


LIME

- Works with any 'black-box' model
- Explanations are only locally-faithful (need not be globally)

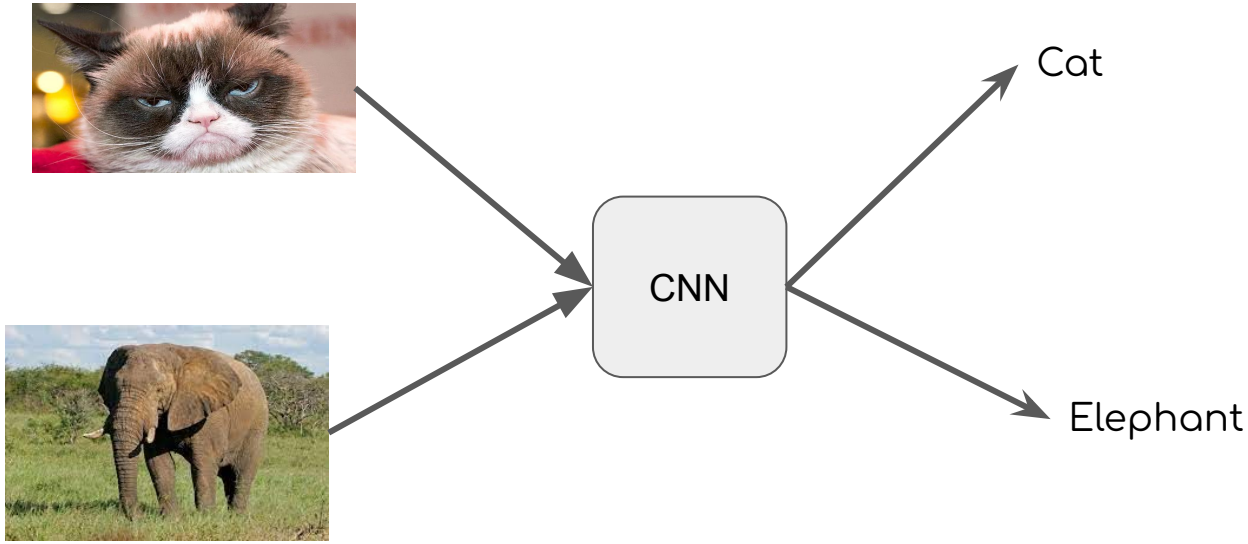


Visual explanations for the DNN based recognition/classification

DNNs - complex ML systems

- DNNs is a success story
- However, they are complex models
- 10s of layers, 100s of activations, 100000000 of parameters

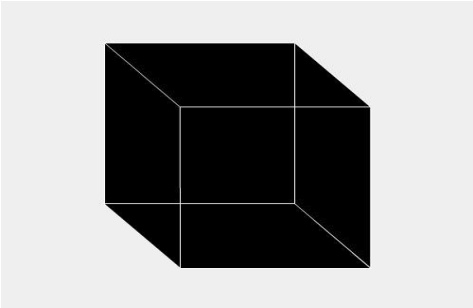
DNN classifier training



DNNs - what do they learn ?

DNNs are black boxes ?

- Often don't provide detailed information about the inference



We need explanation

- Some sort of reasoning for the prediction
  - Users could relate the i/p and the predicted o/p

E.g. Visual Explanation

DNN Explanations

Neuron Visualization

Evidence Localization

Feature Reconstruction

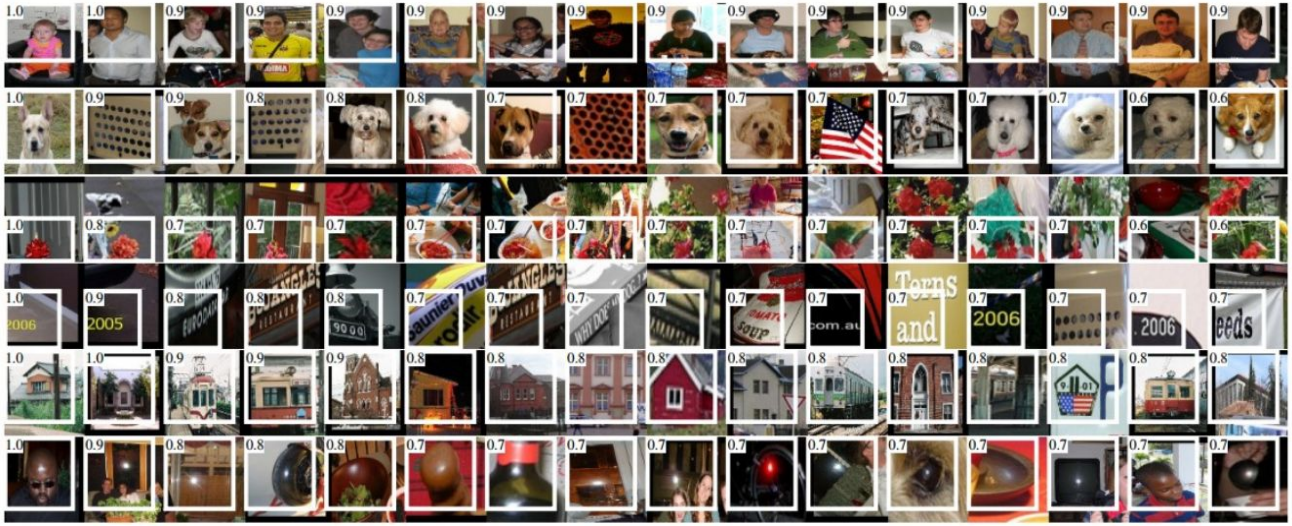
Neurons and stimuli

- What do the neurons learn ?

Ross Girshick et al. CVPR 2014

Neurons and stimuli

- Recognize visual attributes/concepts/topics



Ross Girshick et al. CVPR 2014

Explanation

Neuron Visualization

Evidence Localization

Feature Reconstruction

Evidence localization

- Provide visual explanations
- Grounding the inference
- Ex: Classification network
  - Which pixels are responsible to the predicted label ?

Class model visualization

- Numerically generate an image for chosen class

$$\arg\max_{I} S_c(I) - \lambda \|I\|_2^2,$$

Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps [S Karenyan et al.] [ICLRW 2014]

Class model visualization

- Numerically generate an image for chosen class

$$\arg\max_{I} S_c(I) - \lambda \|I\|_2^2,$$

dumbbell cup dalmatian

Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps [S Karenyan et al.] [ICLRW 2014]

Image specific visualization

- Query the CNN about the spatial support for a class
- Compute the gradients wrt the image

Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps [S Karenyan et al.] [ICLRW 2014]

Image specific visualization

- Equivalent to performing gradient ascent on score function wrt image

$$S_c(I) \approx w^T I + b \qquad w = \left.\frac{\partial S_c}{\partial I}\right|_{I_0}$$

Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps [S Karenyan et al.] [ICLRW 2014]

Image specific visualization

Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps [S Karenyan et al.] [ICLRW 2014]

Class activation maps

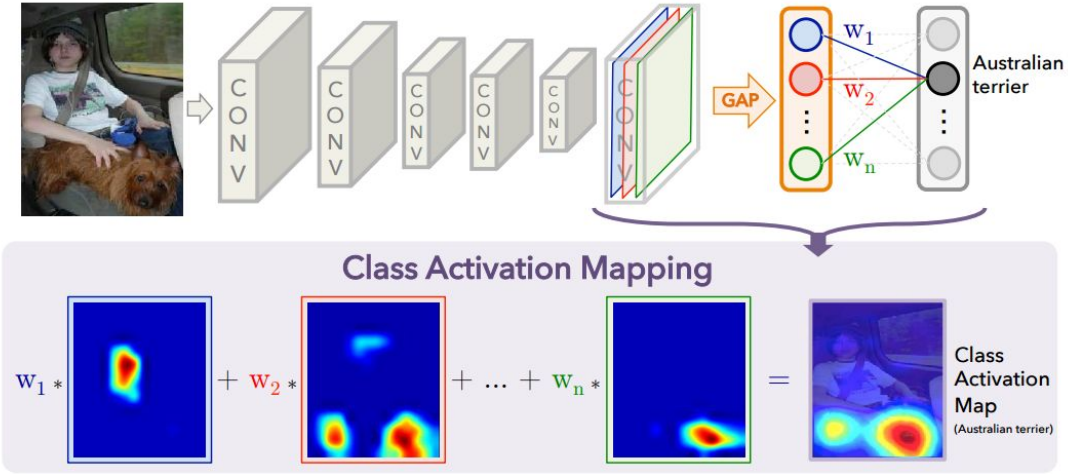
Class Activation Maps (CAM)

- Class discriminative image regions used by CNN to identify the category

B Zhou et al. Learning Deep Features for Discriminative Localization, CVPR 2016

Class Activation Maps (CAM)

- Perform GAP on the final conv feature map

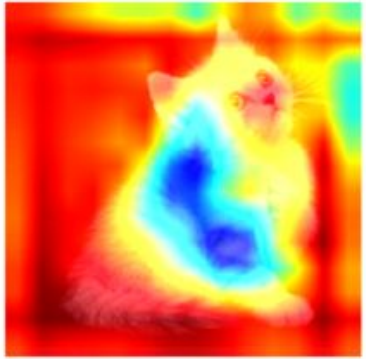


B Zhou et al. Learning Deep Features for Discriminative Localization, CVPR 2016

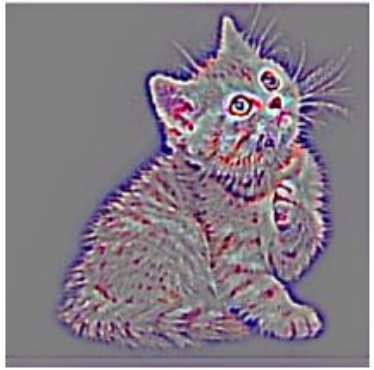
Pixel Attributions

Original Image

Grad CAM



Guided Backpropagation



Guided Grad CAM

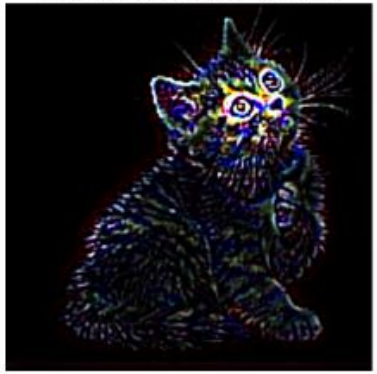


Figure from Renu Khandelwal and towardsdatascience.com

Visualization

Neuron Visualization

Evidence Localization

Feature Reconstruction

Inverting deep representations

Understanding deep image representations by inverting them

Mahendran et al. CVPR 2015

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भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

DiL
Data-driven Intelligence & Learning Lab

Inverting deep features

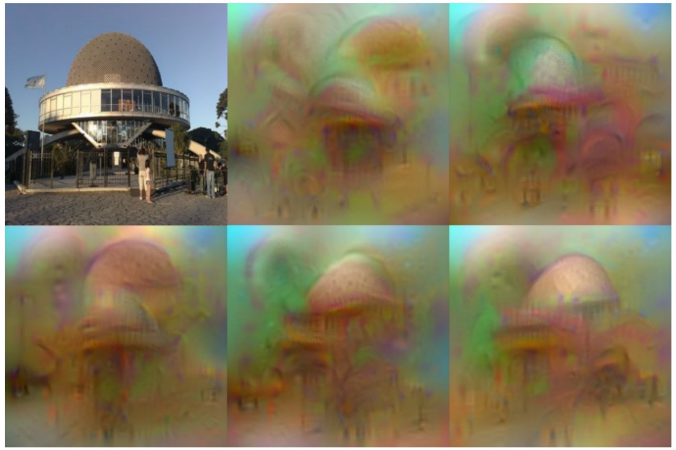


Figure 1. **What is encoded by a CNN?** The figure shows five possible reconstructions of the reference image obtained from the 1,000-dimensional code extracted at the penultimate layer of a ref-

Inverting deep features

- Given an image encoding, to what extent we can reconstruct the image ?
- No unique solution

$$\mathbf{x}^* = \underset{\mathbf{x} \in \mathbb{R}^{H \times W \times C}}{\operatorname{argmin}} \ \ell(\Phi(\mathbf{x}), \Phi_0) + \lambda \mathcal{R}(\mathbf{x})$$

Loss function & Regularizer

$$\ell(\Phi(\mathbf{x}), \Phi_0) = \|\Phi(\mathbf{x}) - \Phi_0\|^2$$

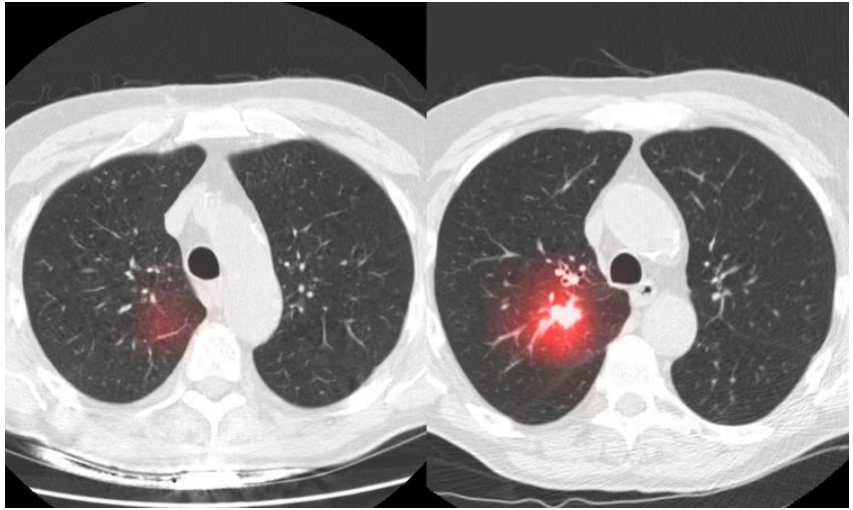
- R - restrict the reconstruction to natural images
- Challenge : modelling it → TV norm prior

$$\mathcal{R}_{V^\beta}(\mathbf{x}) = \sum_{i,j} \left((x_{i,j+1} - x_{ij})^2 + (x_{i+1,j} - x_{ij})^2 \right)^{\frac{\beta}{2}}$$

Results

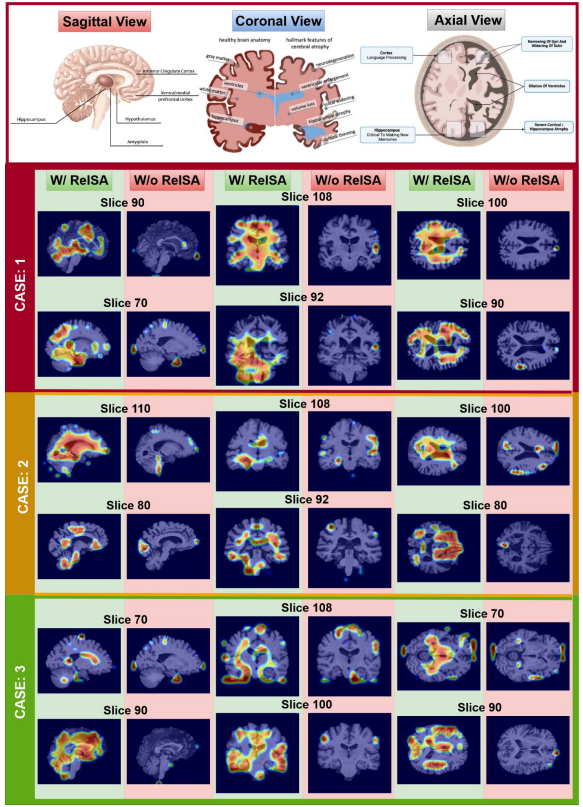
conv1
relu1
mpool1
norm1
conv2
conv4
relu4
conv5
relu5
mpool5
relu2
mpool2
norm2
conv3
relu3
fc6
relu6
fc7
relu7
fc8

Lung cancer detection & explanation



CT scans show the accuracy of Sybil -- the AI tool developed by MIT and Mass Gen -- at detecting early lung cancer. The left image shows what Sybil detected in a patient (highlighted in red) and the right image of the same patient shows what radiologists detected two years later (highlighted in red).

Alzheimer's Diagnosis & Explanation



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Summary

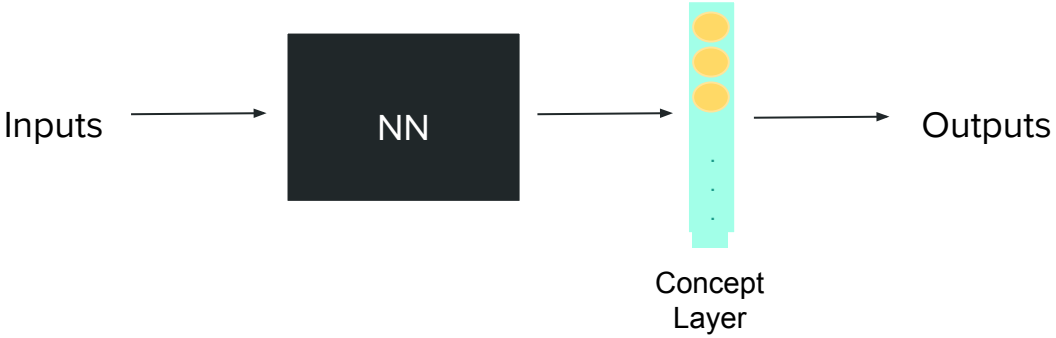
- Transparency is useful at three different stages of Artificial Intelligence (AI) evolution
- AI is significantly weaker than humans - not yet reliably 'deployable' (e.g, QA)
    - Identify the failure modes
- AI is on par with humans - reliably 'deployable' (e.g, recognition)
    - To establish appropriate trust and confidence in users
- AI is significantly stronger than humans - e.g, Chess/Go
    - Machine teaching

Advanced Explainability Frameworks in Vision

Concept Bottleneck Models

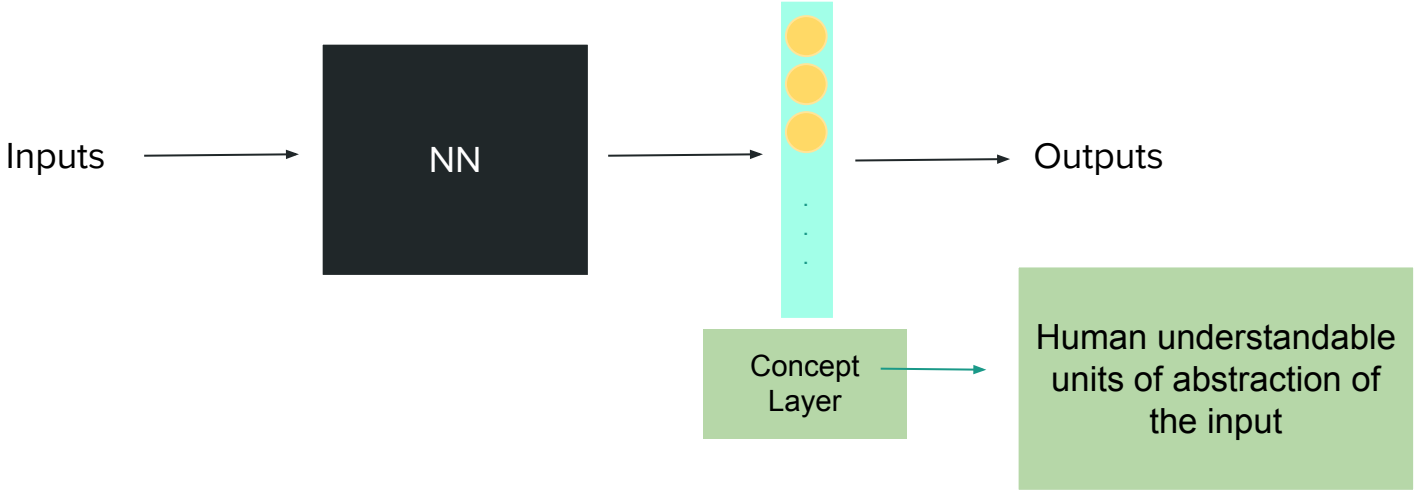
- Standard Classifiers

- Concept Bottleneck Models



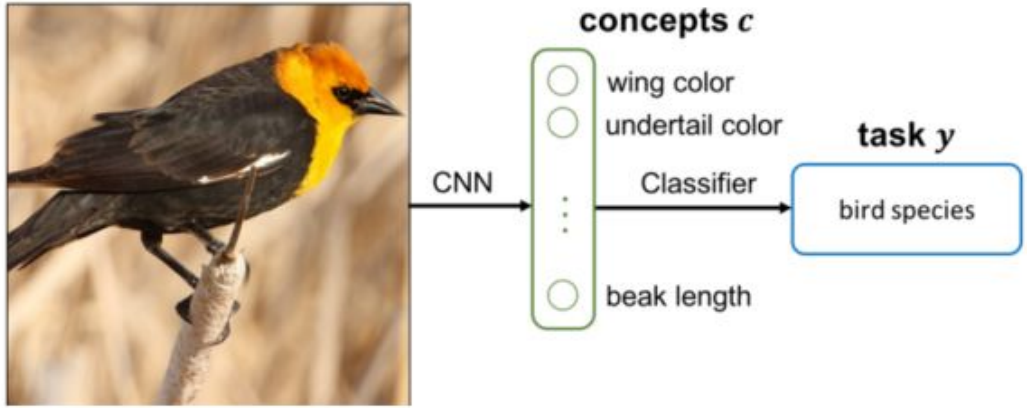
Concept Bottleneck Models

- Learn to predict a set of user provided concepts → Use concepts for downstream task

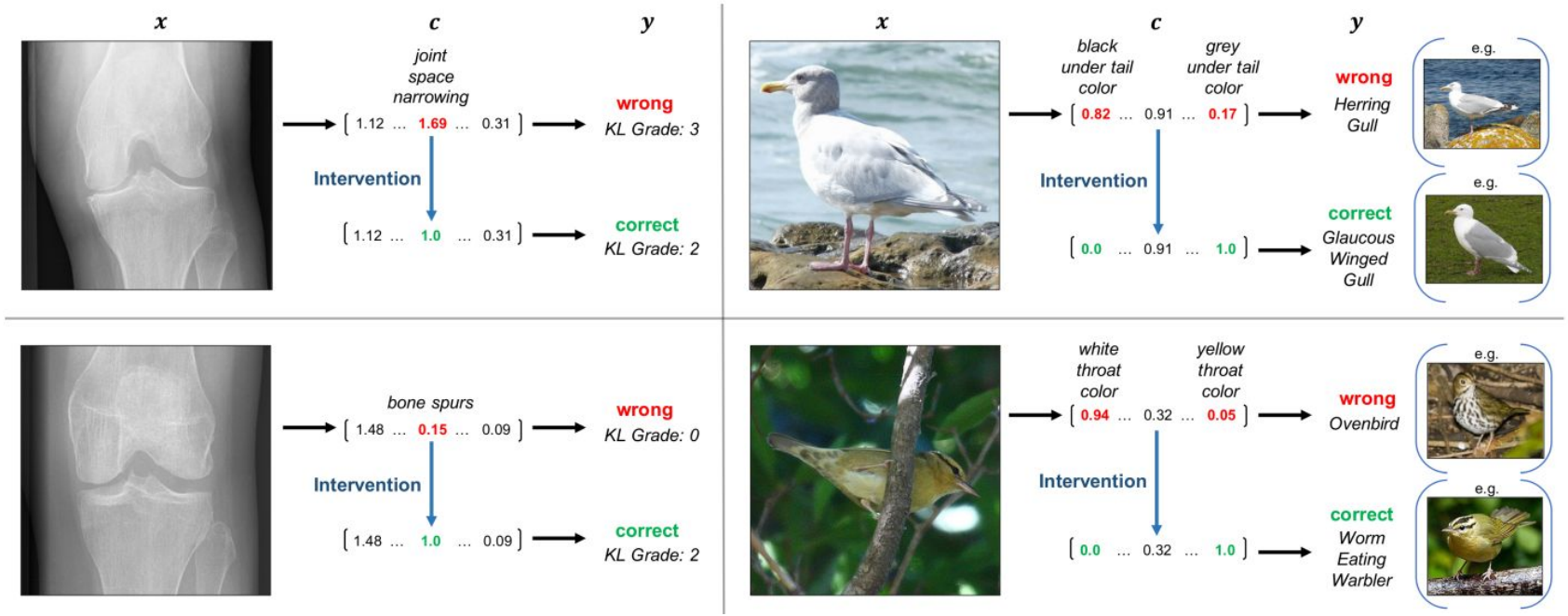


Concept Bottleneck Models

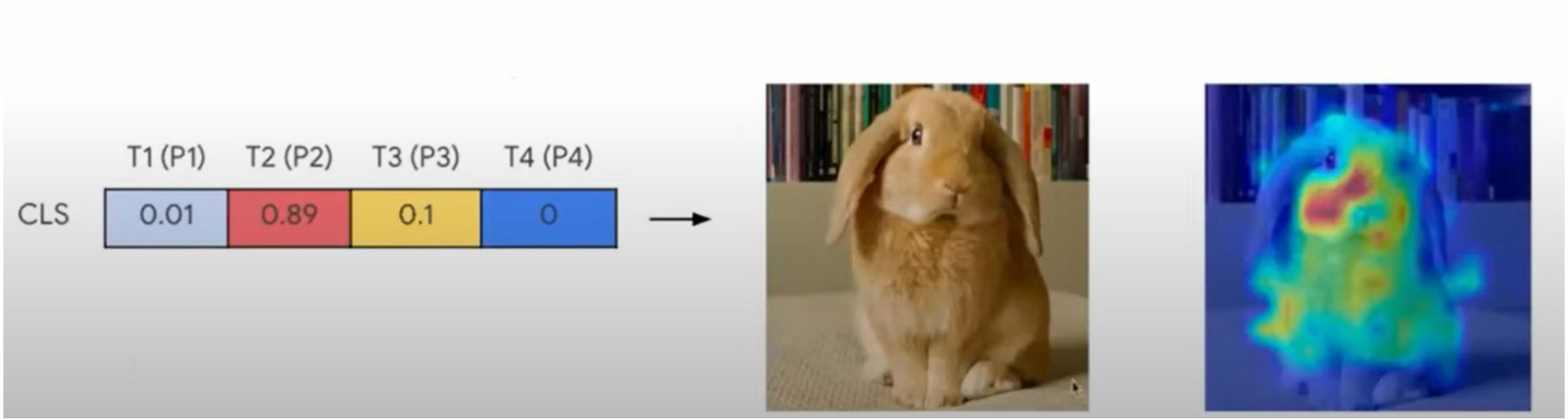
- Neurons in the bottleneck layer → concepts
- Predicted class ← (Usually Linear) combination of concepts



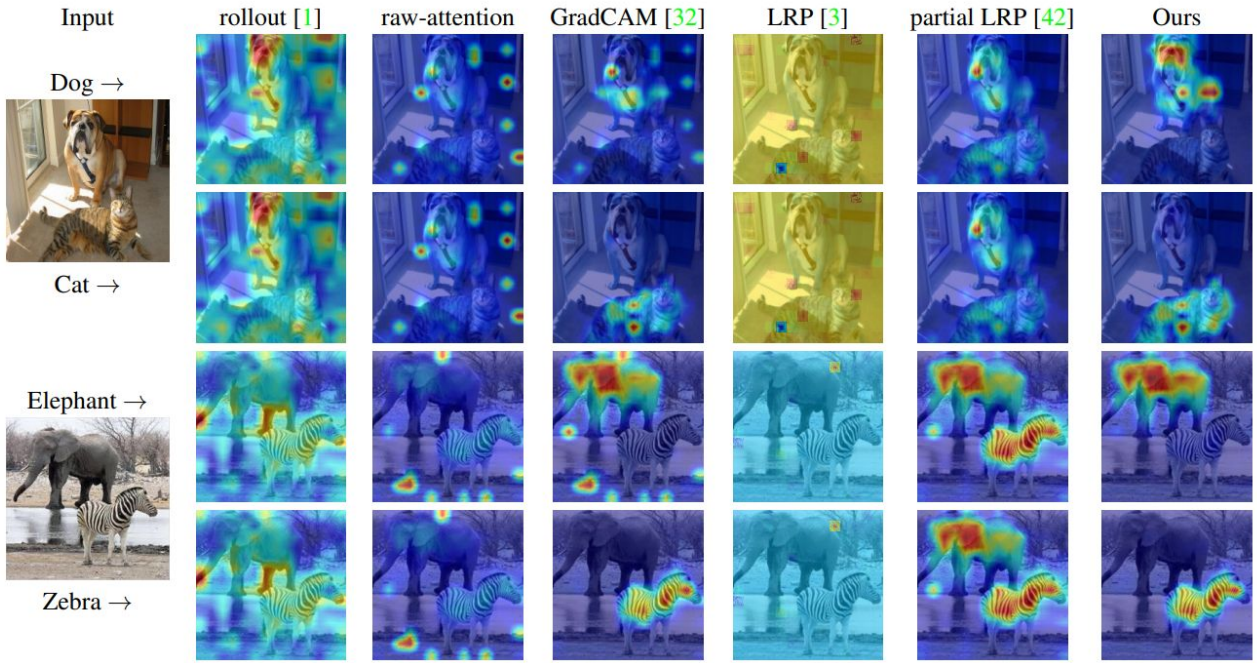
Test-time Interventions



Is attention in ViTs an explanation?



TiBA [CVPR 2021]



Explainability in LLMs

Stochastic parrots(?)

- ‘LLMs though able to generate plausible language, do not understand the meaning of the language they process’ [Bender et al. 2021]

Figure credit: eight2late.wordpress

Reasoning vs. Mimicry

- Sophisticated mechanism to predict the next word
- LLMs mimic
  - Basically, look for the words that fit together

No consideration for the verification

- LLMs have no access to structured, factual knowledge
  - No way for them to check (other sources)

Not very conservative

- Not optimized to refuse to answer (frequently)
  - E.g. ‘I don’t know’ or ‘I am not sure’ responses

Prompt:

```
I went to the market and bought 10 apples.
I gave 2 apples to the neighbor and 2 to the repairman.
I then went and bought 5 more apples and ate 1.
How many apples did I remain with?
```

Output:

```
11 apples
```

Prompt:

```
I went to the market and bought 10 apples.
I gave 2 apples to the neighbor and 2 to the repairman.
I then went and bought 5 more apples and ate 1.
How many apples did I remain with?
Let's think step by step.
```

Output:

```
First, you started with 10 apples.
You gave away 2 apples to the neighbor and 2 to the repairman, so you had 6 apples left
Then you bought 5 more apples, so now you had 11 apples.
Finally, you ate 1 apple, so you would remain with 10 apples.
```

CoT Prompting

Going forward - Retrieval to Reasoning

- Causality vs Correlations
- Multi-modal explanations
- Quality of explanations
- Model-Agnostic (generic) explanations
- User-centric explanations
- Scalable and Robust explanations
- etc.

THANK YOU